

Operation and stability of an emissive cathode in a high density plasma



Francis Pagaud, Vincent Dolique, Nicolas Plihon
ENSL, CNRS, Laboratoire de Physique, F-69342 Lyon, France



OBJECTIVE

A tungsten emissive cathode is used to control plasma properties such as Φ_p . Its thermionic emission must be known to model the potential drop at the plasma core. Yet the emitted current depends unexpectedly on its bias. The goal is to **understand the emission properties of this cathode in a pre-existing plasma**.

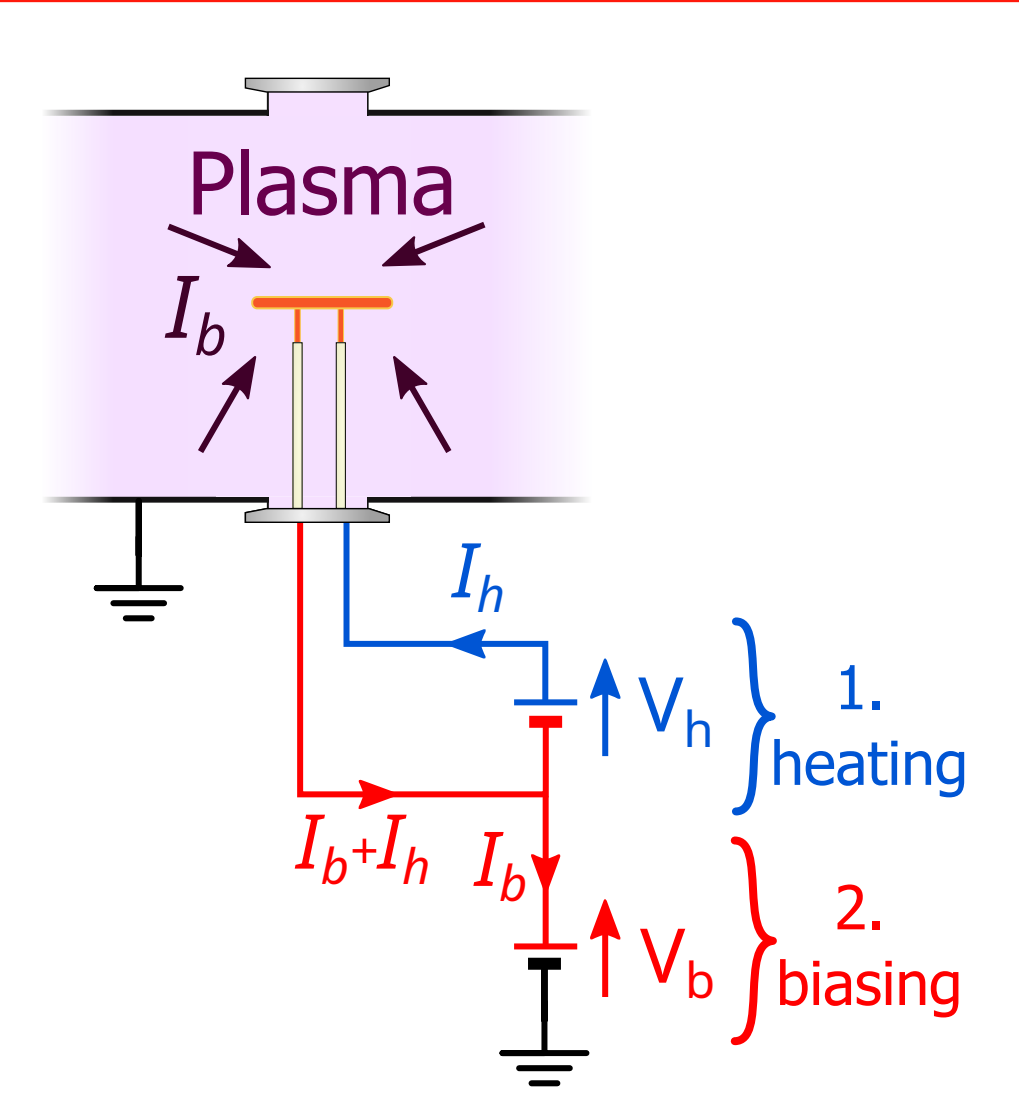
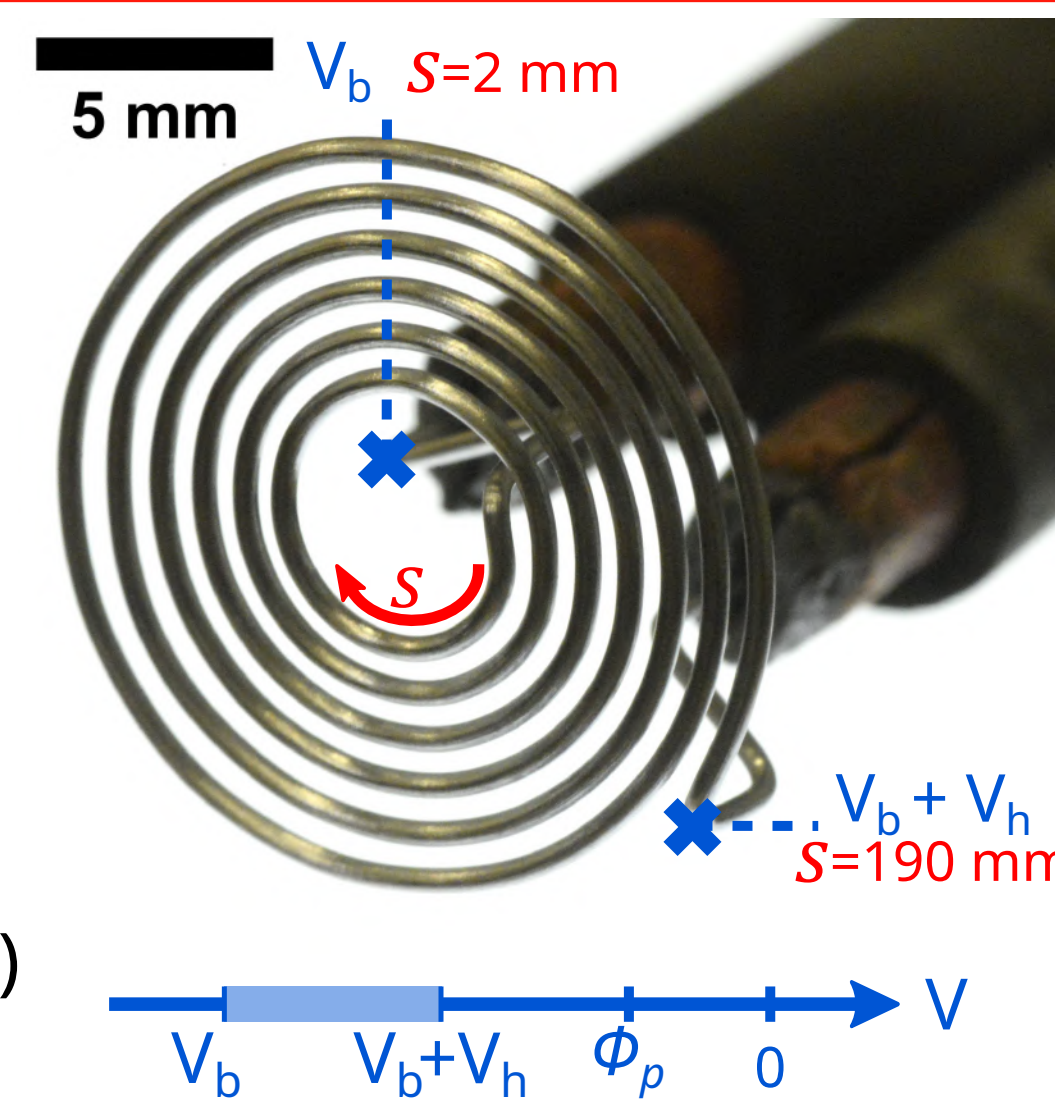
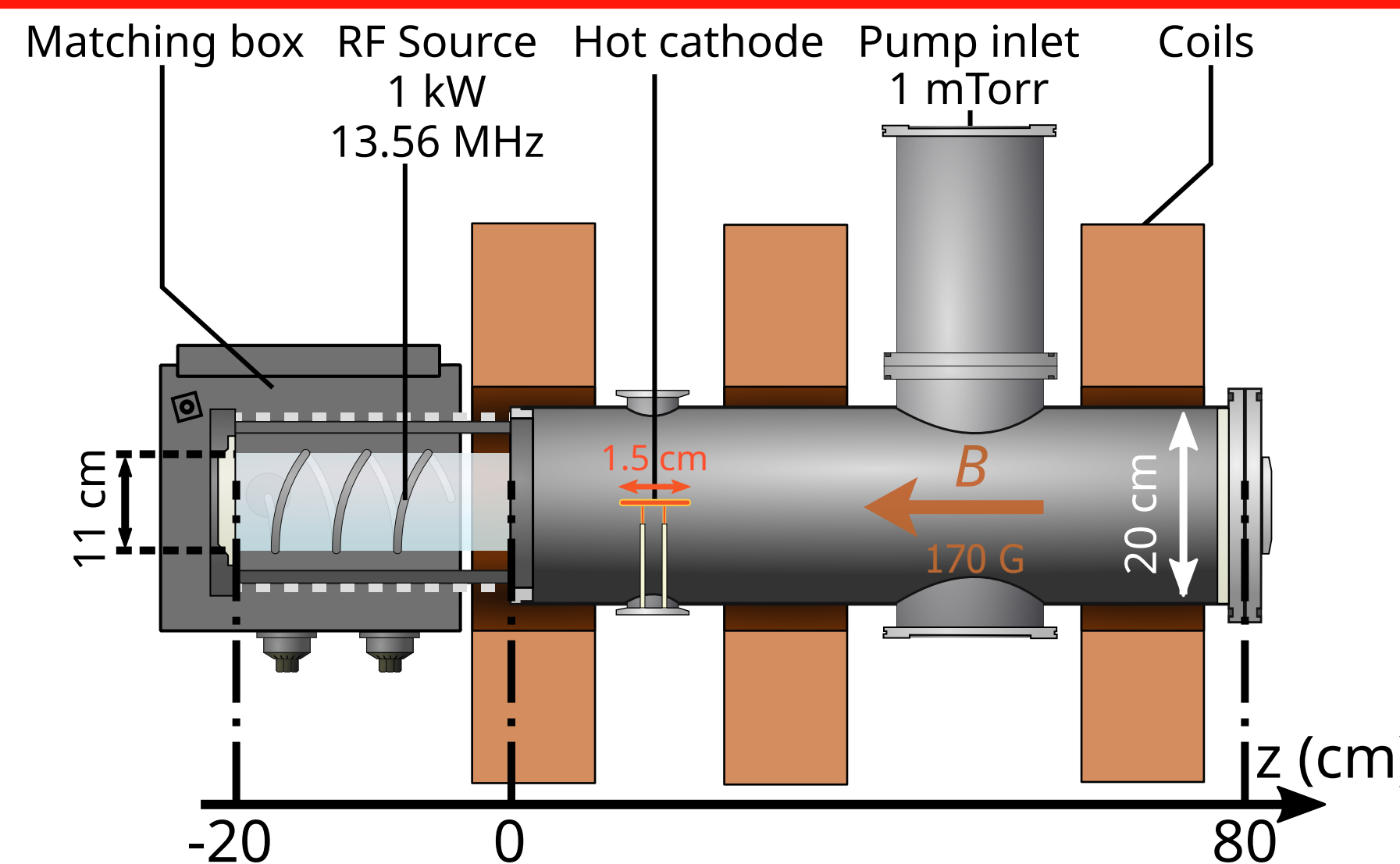
1. Experimental setup

Plasma setup: ▶ RF source (13.56 MHz), magnetized [1].
▶ DC heated tungsten cathode

Video of a plasma shot with a strong emitted beam



Quantity	Pressure	Mag field	RF Power	Density	Electron temperature
Value	1 mTorr	170 G	1 kW	$2 \cdot 10^{18} \text{ m}^{-3}$	3 – 4 eV



2. I_h and V_b coupling

Naive prediction:

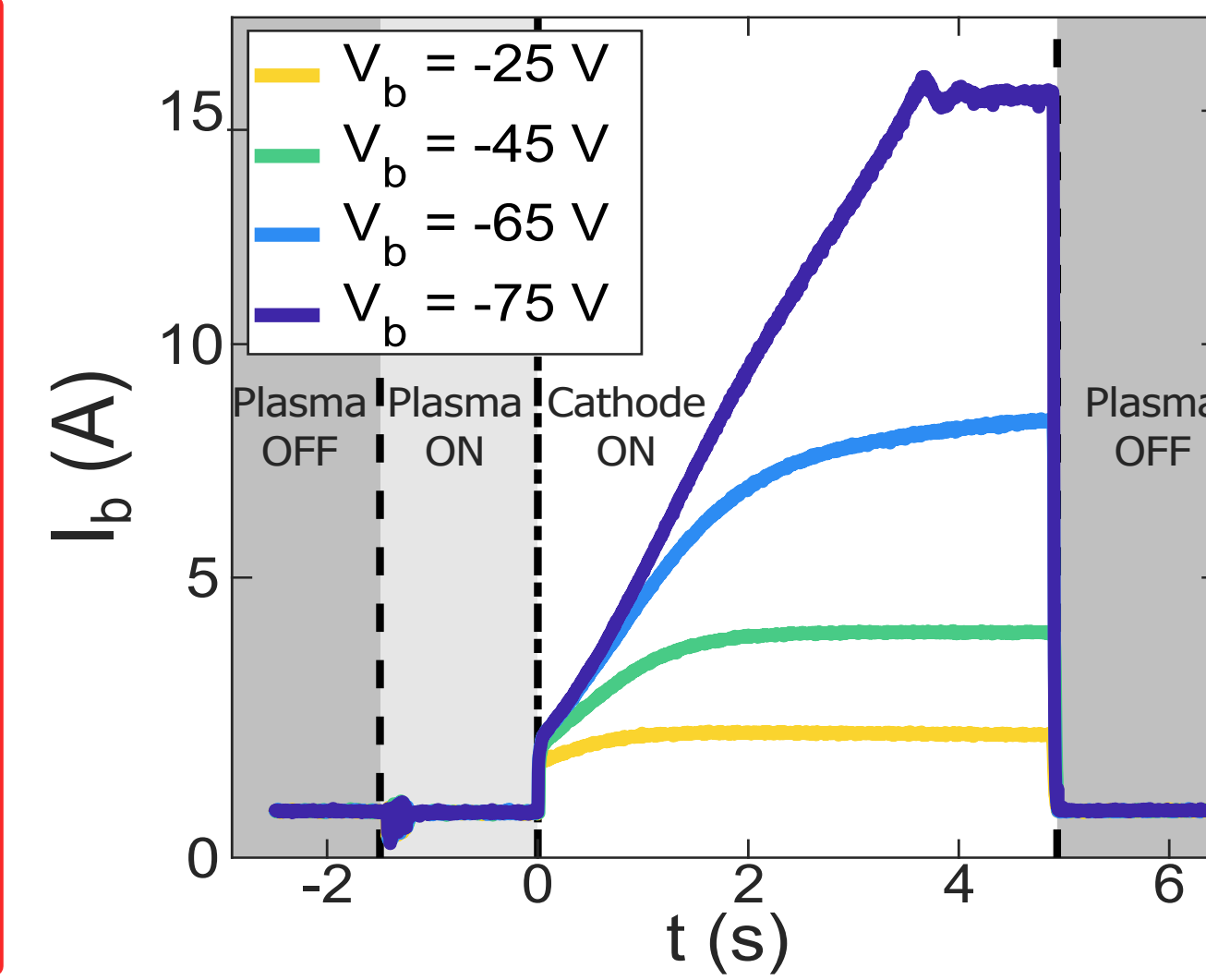
▶ I_b is controlled by I_h independently of V_b .

$$I_b = I_{em} + I_h - I_e$$

$I_i, I_e \ll I_{em}$

$$I_{em} = A_G S T_W^2 \exp\left(-\frac{eW}{k_B T_W}\right)$$

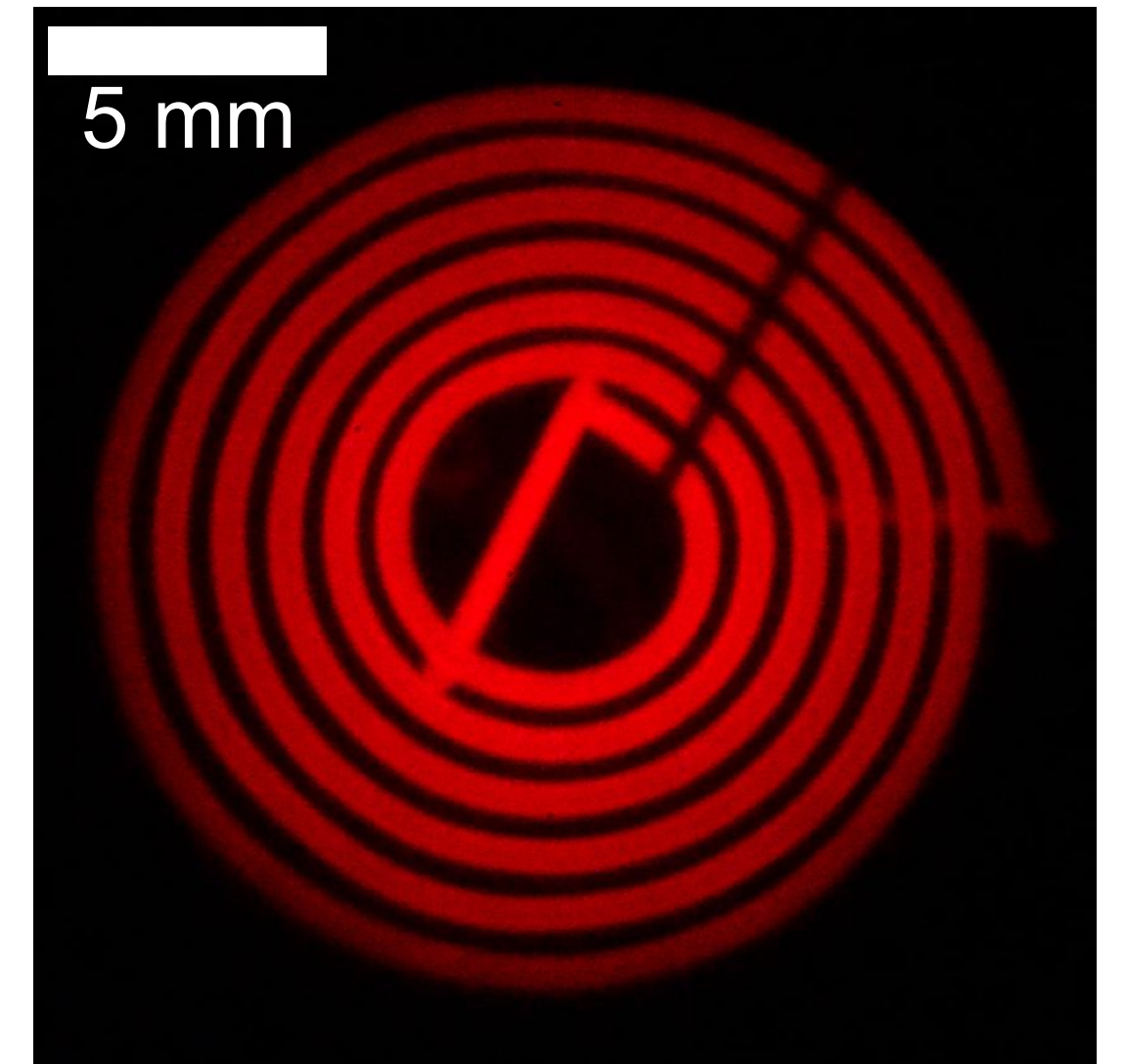
Temperature-driven emission



Experimental results:

▶ Dependence on V_b .
▶ Runaway behaviour.

T_W investigation with an optical pyrometer



3. Spatially and temporally resolved temperature measurements

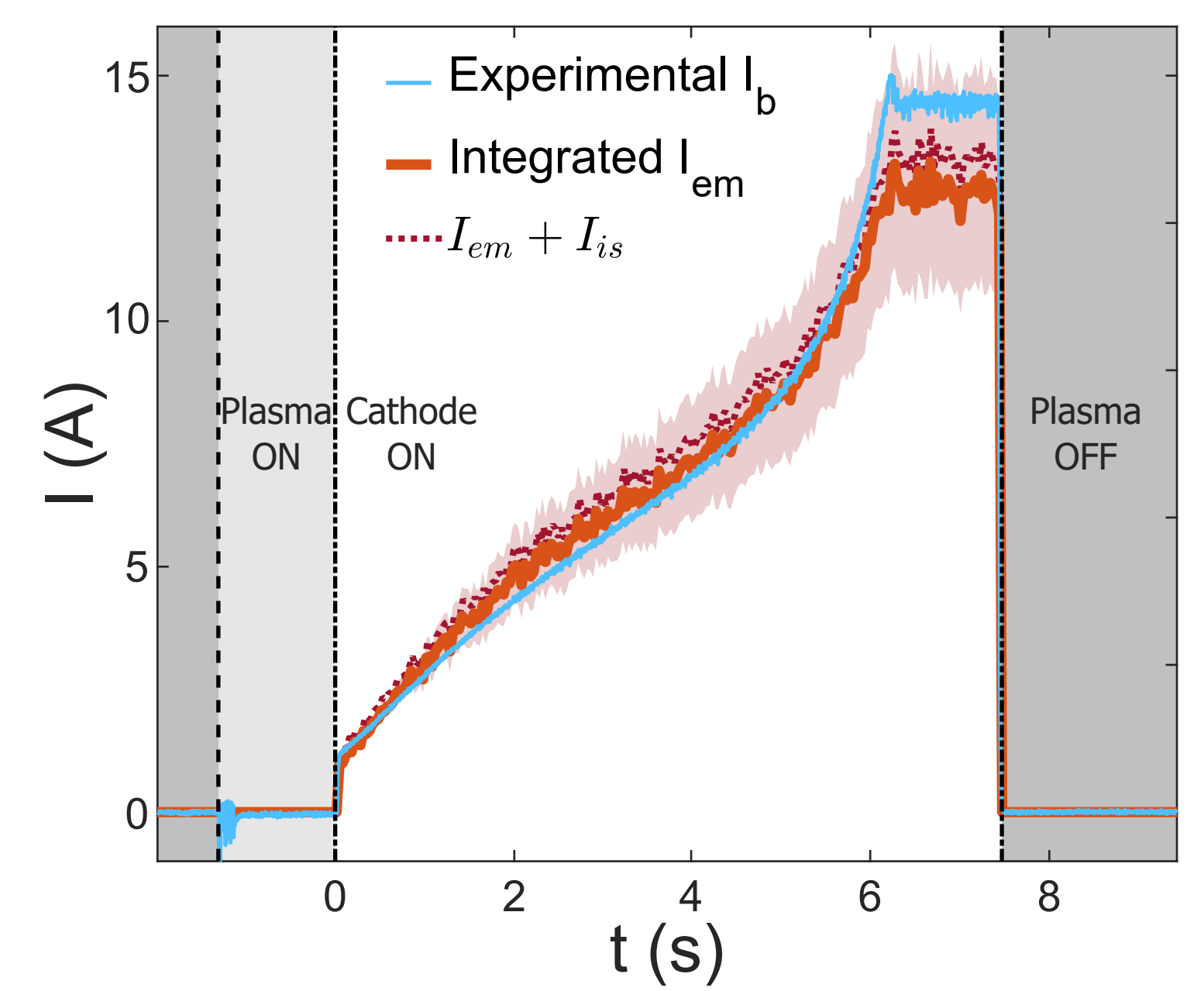
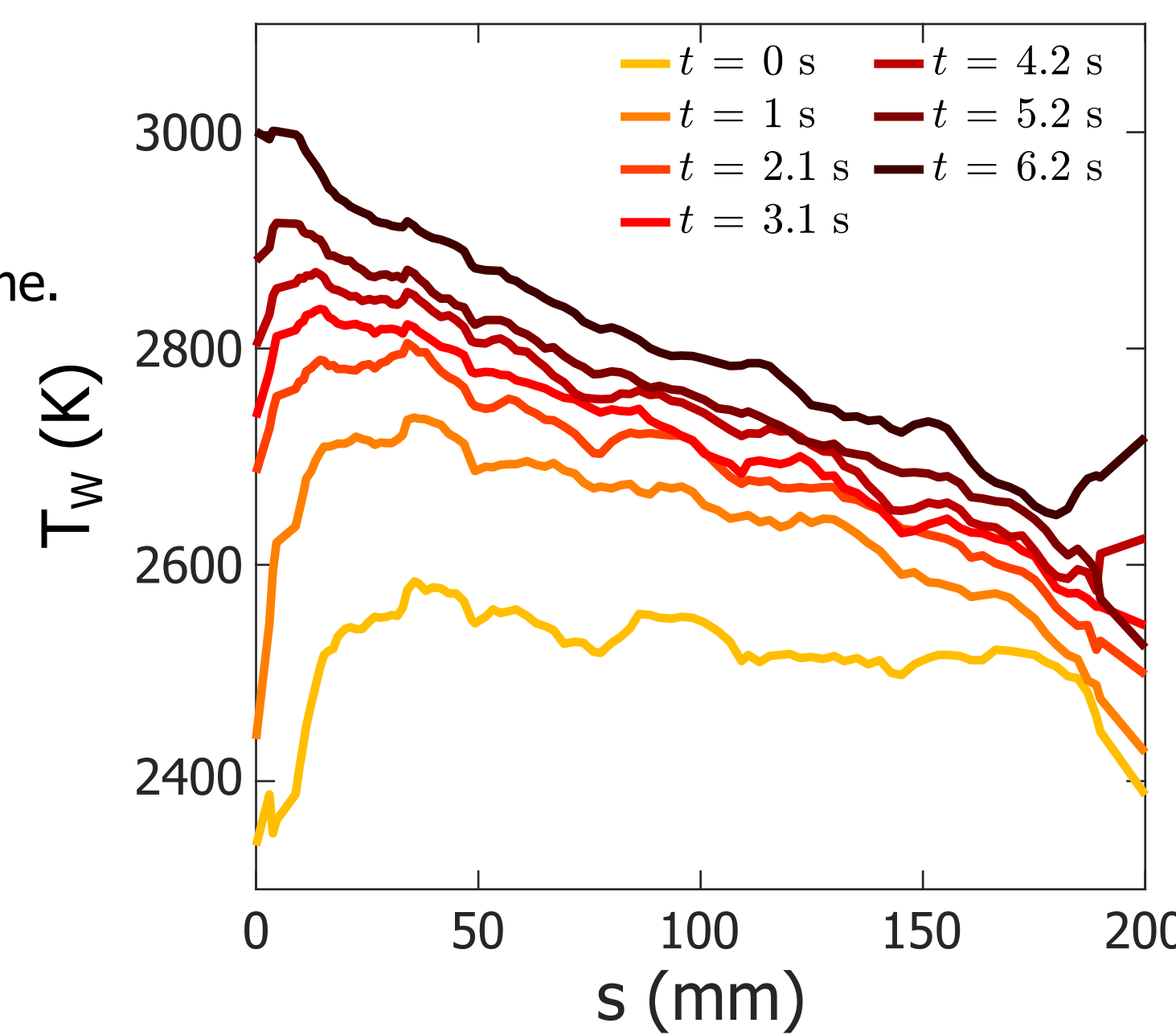
Pyrometer experiment:

1. Light intensity measurement at 650 nm.
2. Conversion to temperature along curvilinear abscissa and time.
3. Computation of thermionic emission.

$$I_{em} = \int_0^s A_G T_W^2 \exp\left(-\frac{eW}{k_B T_W}\right) 2\pi r dx$$

Experimental results:

- ▶ Heterogeneous T_W profile.
- ▶ Good agreement with experimental I_b .



4. Numerical thermal model

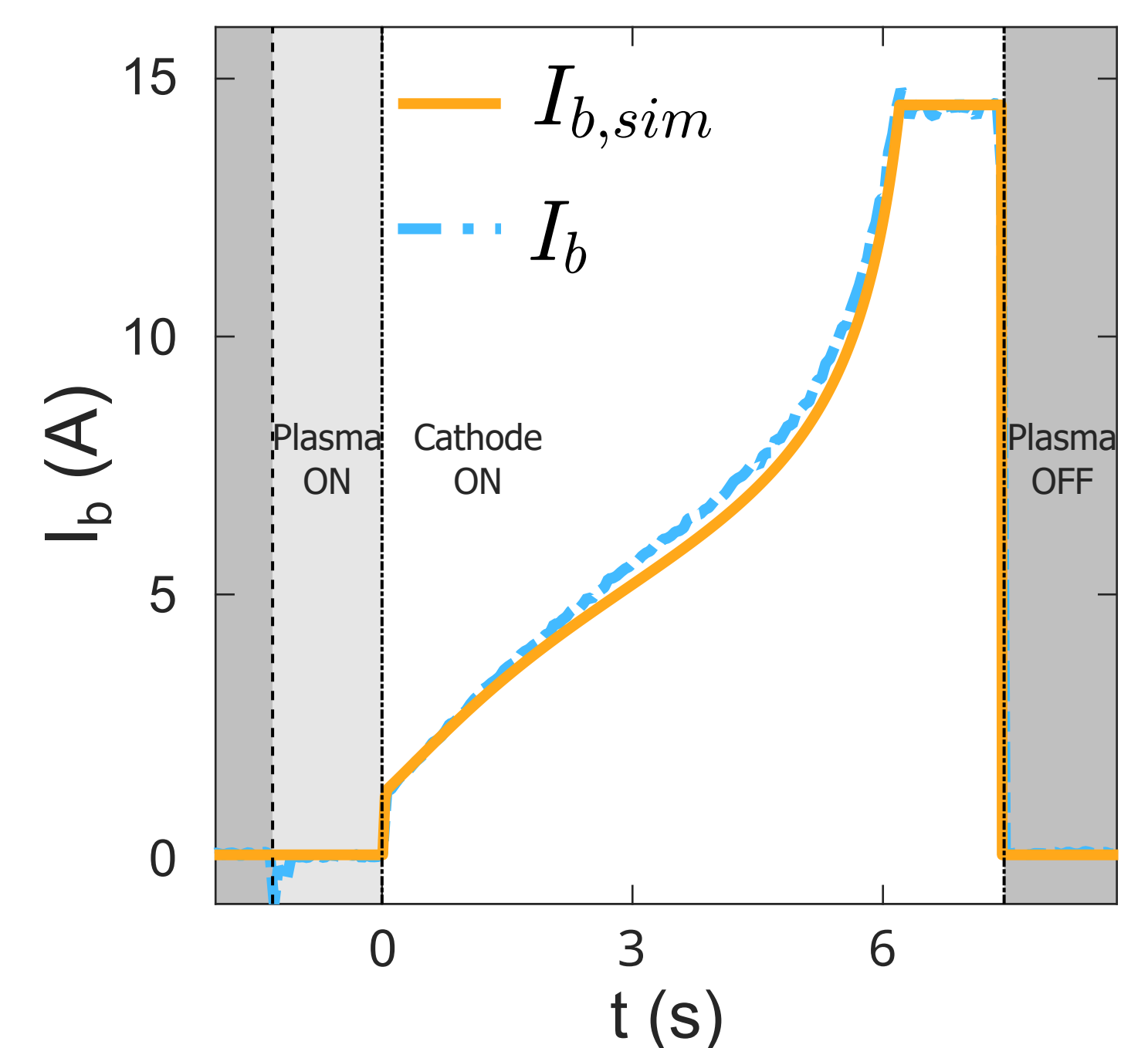
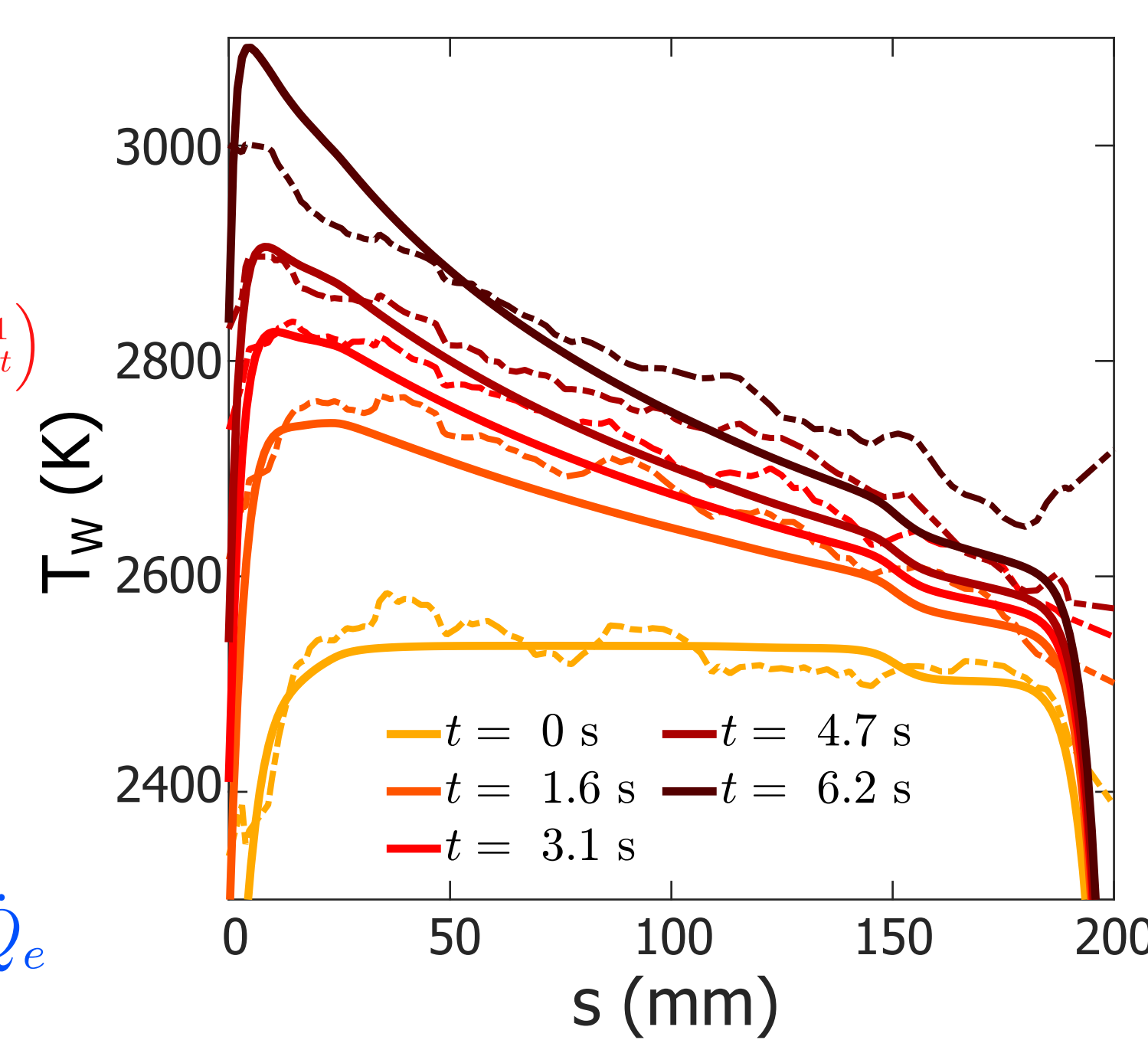
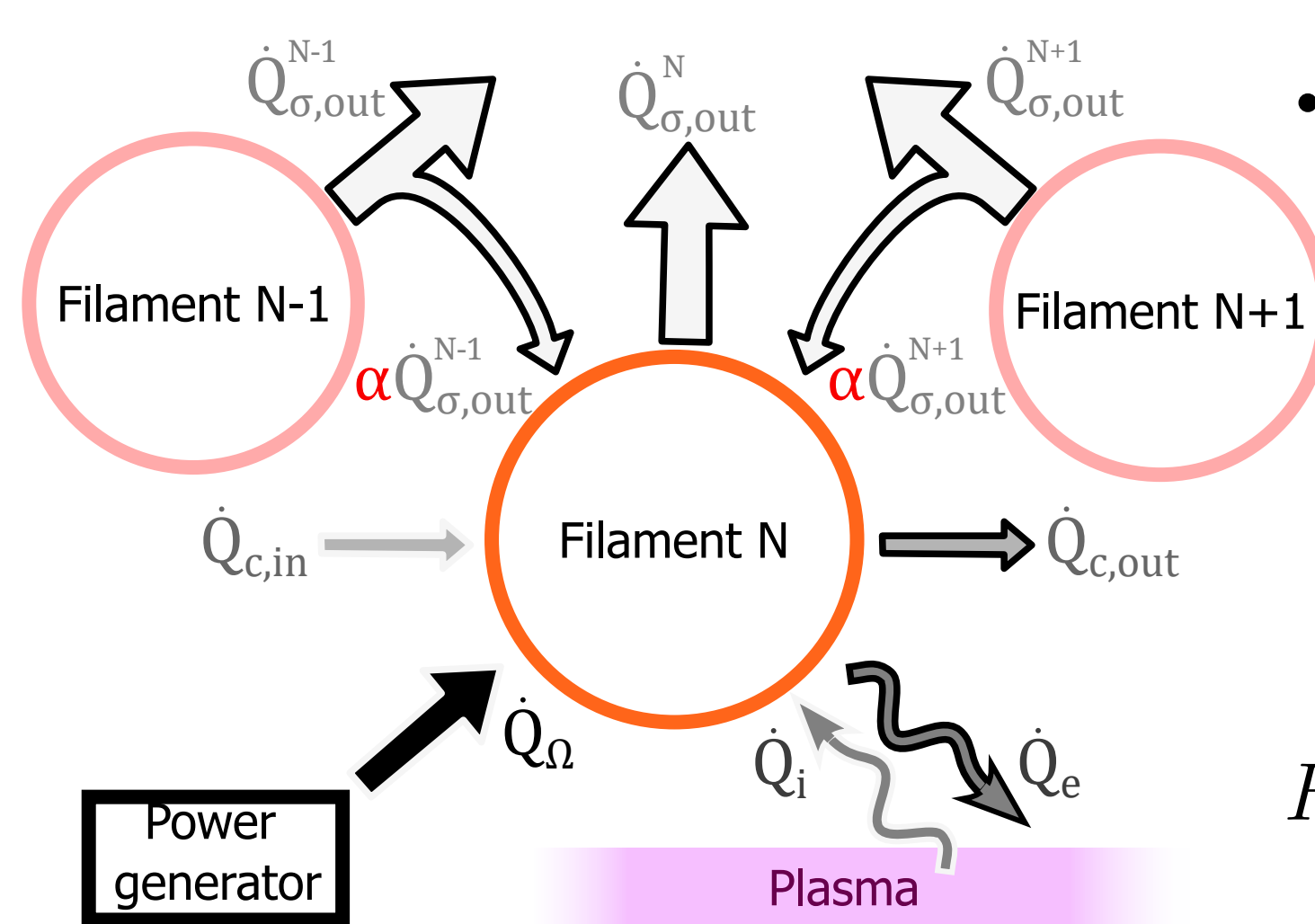
- Thermal inertia $\dot{H} = C_p \rho V \frac{\partial T_W}{\partial t}$
- Conduction $\dot{Q}_c = \lambda \pi r^2 \frac{\partial^2 T_W}{\partial s^2}$

- Ohmic heating $\dot{Q}_\Omega = R \left(I_h + \int_s^{L_W} i_b du \right)^2$
- Outward dissipative radiation $\dot{Q}_{\sigma, out} = \sigma S \epsilon T_W^4$

- Inward dissipative radiation $\dot{Q}_{\sigma, in} = \alpha \epsilon \left(\dot{Q}_{\sigma, out}^{N-1} + \dot{Q}_{\sigma, out}^{N+1} \right)$
- Thermionic cooling $\dot{Q}_e = \frac{I_{em}}{e} (eW + 2k_B T_W)$
- Ion bombardment $\dot{Q}_i = I_i [E_i - W + \beta (\phi_p - V_b)]$

Integro-differential equation

$$\dot{H} = \dot{Q}_c + \dot{Q}_\Omega + \dot{Q}_i + \dot{Q}_{\sigma, in} - \dot{Q}_{\sigma, out} - \dot{Q}_e$$



5. Application to I_b prediction and measurement

I_b prediction and runaway behaviour:

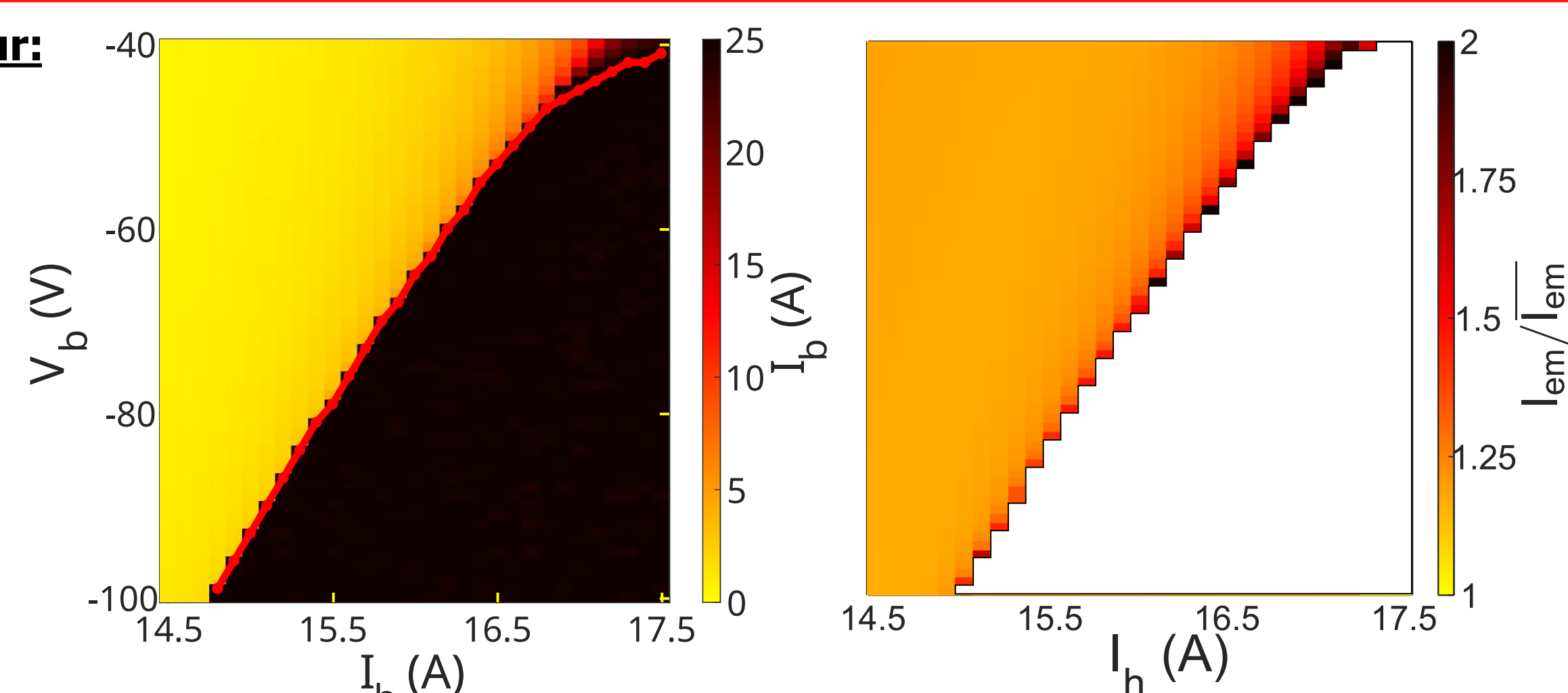
- ▶ Anticipate I_b for different parameters.
- ▶ Understand V_b and I_h dependencies.
- ▶ Avoid damaging strong emission.

Versatile T_W measurement method:

- ▶ Get the global cathode resistance.
- ▶ Deduce $\bar{T}_W$ value through resistivity.

$$\rho_W = a T_W^{1.22} + b$$

- ▶ Correct homogeneous $\bar{T}_W$ assumption using numerical results.



CONCLUSION

A complex plasma-cathode dynamic system has been investigated experimentally and numerically with an excellent agreement.

It provides a fine understanding of the physical phenomena at play as well as new methods to operate this simple cathode. Such emissive cathode could be commonly use to achieve plasma control.